

Shi Yu Regine Wong

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EDUCATION

Digipen Institute of Technology, Singapore | Singapore

Sep. 2024 – Sep. 2028

B.S. in Computer Science in Real-Time Interactive Simulation | GPA: 3.7/4.0

- **Relevant Courses:** Files and Databases, Modern Web Development, Mobile Operating Systems, Computer Organization, Data Structures, Algorithms & Analysis, Foundations of Programming
- **Honors:** Dean's List, Honors

Temasek Polytechnic | Singapore

Apr. 2020 – Apr. 2024

Diploma in Aerospace Electronics | GPA: 3.5/4.0

- **Relevant Courses:** Computing & Programming for Problem Solving, Circuit Analysis, Data Visualisation and Analytics, and Basic Aerodynamics
- **Honors:** top 25% of cohort, in Polytechnic Foundation Programme

Boot.dev | Singapore

Aug. 2026 – Present

- Successfully completed Boot.dev, a rigorous backend engineering curriculum.
- Finished over 23 courses and projects, including courses on DSA, functional programming, web servers, and SQL databases.

TECHNICAL SKILLS

Languages: C++, Python, C, Golang, SQL

Tools and Frameworks: Git, Docker, VSC

AI & Data Tools: Claude API, OpenAI API, GitHub Copilot

Operating Systems: Windows, Linux

PROJECTS

Real-Time Graphics Engine | DigiPen CSD2150 (Computer Graphics)

Jun. 2026

- Built a C++/OpenGL rendering pipeline featuring ray tracing, deferred shading, and HDR cubemap environment mapping (reflection/refraction) across a custom multi-window GLFW application
- Developed a GPU compute-shader raycaster and multi-pass post-processing effects (blur, edge detection, cartoon shading, portal rendering), unifying a semester's worth of graphics assignments into one portfolio demo

Rock & Ghoul | DigiPen Team Game Project

Apr. 2026

- Architected a custom C++ game engine (CoreEngine) around a hashmap-based ECS with 14 dependency-enforced component types, and an integrated ImGui/ImGuizmo level editor
- Built ImGuizmo-driven transform/rotate/scale gizmo with hover/active-state gating to prevent conflicts with surrounding editor UI
- Implemented a command-pattern undo/redo system with a 30-action stack, covering entity creation/deletion, gizmo transform edits, and generic component-property edits via lambda-based apply/undo functions
- Built a state-machine sprite animation system driving per-entity animation switching (idle/walk/attack/hit/death) across enemy and player controllers

RELEVANT EXPERIENCE

Rolls-Royce@NTU Corporate Laboratory | Singapore

Apr 2023 – Aug 2023

Internship

- Acquired displacement, voltage, and temperature data from piezoelectric actuator test rigs using NI and Hioki data acquisition systems
- Developed MATLAB scripts to process and analyze actuator test data (e.g., filtering noise, plotting displacement/frequency response)
- Conducted temperature testing using thermocouple acquisition such as National Instruments (NI) and Hioki
- Conceptualized wave forms using ARB_EDIT

LEADERSHIP EXPERIENCE

DigiPen Institute of Technology Singapore | Singapore

Aug. 2026 - Present

crunchyPeanut Technical Lead

- Led a 6-engineer team building AirDraw, a gesture-controlled 3D presentation tool (Unity/C#/MediaPipe), setting engine architecture, assigning work across 9 deliverables, and personally owning the hand-tracking input service and 3D session replay system.

DigiPen Institute of Technology Singapore | Singapore

Jan. 2026 - Apr. 2026

AvidDivas Technical Lead

- Led a team of 8 engineers on Rock & Ghoul, directing engine architecture and system design decisions and coordinating work across editor tooling, gameplay, and animation systems